SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CAPSTONE PROJECT SUBMISSION PORTFOLIO

OrderPulse: Food Order Performance Analytics

A Three-Tier Snowflake Medallion Data Engineering & Streamlit Analytics Platform

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Capstone Track:	Snowflake Data Engineering (SmartBridge / MySkillWallet.ai)
Cohort Name:	Snowflake Data Engineering_EL_2027_Cohort 1
GitHub Repository:	https://github.com/ArnaovGarg925/OrderPulse-Food-Order-Performance-Analytics
Demonstration Video:	https://drive.google.com/file/d/12-2-9LxKNoYeqRWb3uGjThnOJETl7Rg3/view?usp=drive_link
Submission Format:	**Official Consolidated PDF (Phases 1 to 8)**

Table of Contents / Capstone Phases

Phase	Phase Template Name	Deliverable Scope
Phase 1	**Template 1: Ideation Phase**	Problem Statement, Objectives & Medallion Overview
Phase 2	**Template 2: Requirement Analysis**	Functional/Non-Functional Specs & Personas
Phase 3	**Template 3: Project Design Phase**	Star Schema ERD, Fact/Dims & Persona Marts
Phase 4	**Template 4: Project Planning Phase**	WBS, 8 Epics / 20 Tasks & Sprint Timeline
Phase 5	**Template 5: Project Development Phase**	Snowflake DDL/DML & Streamlit App Architecture
Phase 6	**Template 6: Performance Testing**	Data Volume, 6.5x Pruning & 24 Engineered Fields
Phase 7	**Template 7: Project Documentation**	GitHub Structure, Execution Guide & Schemas
Phase 8	**Template 8: Project Demonstration**	1080p Video Walkthrough Agenda & Key Insights

CAPSTONE PROJECT SUBMISSION TEMPLATE

Template 1: Ideation Phase

Project Title:	**OrderPulse: Food Order Performance Analytics**	**Template Phase:**	**Template 1: Ideation Phase**
Student Name:	Arnaov Garg	**Email:**	ag5088@srmist.edu.in
Institution:	SRM Institute of Science and Technology (SRMIST)	**Track / Cohort:**	Snowflake Data Engineering (Snowflake Data Engineering_EL_2027_Cohort 1)
GitHub Repo:	https://github.com/ArnaovGarg925/OrderPulse-Food-Order-Performance-Analytics	**Demo Video:**	Google Drive (1080p Video Live)

1. Project Title & Domain

Title: OrderPulse: Food Order Performance Analytics

Domain: Cloud Data Warehousing, E-Commerce Analytics & Quick-Commerce Logistics

Target Platform: Snowflake Cloud Data Platform & Streamlit Web Applications

2. Problem Statement

Modern online food ordering platforms operate at hyper-scale, capturing millions of data points across diners, restaurant partners, multi-course menus, line-item baskets, delivery agents, and financial settlements. However, in transactional operational environments, this rich telemetry is isolated across disparate silos (e.g., MySQL ordering databases, restaurant partner portals, customer CRM records, and third-party delivery dispatchers).

Because these disparate sources are not integrated into an unified analytical model, business decision-makers face severe challenges: (1) Discrepancy between Gross Merchandise Value (GMV) and realized net revenue due to unclassified delivery cancellations; (2) Blind spots in customer retention, making it impossible to separate one-time buyers from high-value repeat loyalists; (3) Subjective partner negotiations lacking empirical evidence connecting restaurant star ratings with customer order volume; and (4) Delivery SLA compliance failures during meal peak windows that go undetected until customer churn occurs.

3. Proposed Solution & Architecture Overview

OrderPulse solves this fragmentation by implementing an enterprise-grade cloud data analytics platform powered by **Snowflake's Medallion Data Architecture** (RAW ➔ STAGING ➔ CURATED):
• **RAW Layer:** Automated ingestion of operational CSV dumps into schema-permissive landing tables via internal stages.
• **STAGING Layer:** Rigorous data cleansing, strict type-casting, string hygiene, and financial outcome classification.
• **CURATED Layer:** High-performance dimensional Star Schema (2 Fact Tables and 5 Dimension Tables) with surrogate key hashing, complemented by persona-driven analytical data marts for instant consumption.
• **Serving Layer:** An interactive 5-tab **Streamlit Web Application** providing real-time KPI scorecards, dynamic filtering, and interactive Plotly visualizations for stakeholders.

4. Core Objectives

- **Single Source of Truth:** Ingest and reconcile 12,000 orders and 32,921 line items across 7 metropolitan Indian cities.
- **Dimensional Analytics:** Design and implement surrogate-key dimensional models with RFM customer loyalty tiers and rating brackets.
- **Executive Financial Reconciliation:** Clearly differentiate between placed GMV (■96.22 Lakhs) and realized net revenue (■89.38 Lakhs).
- **Persona-Based Decision Support:** Equip Priya (Customer Engagement), Daniel (Restaurant Partnerships), and Grace (Executive Strategy) with tailored analytical marts.

- **Performance Benchmarking:** Achieve sub-second query latency and demonstrate over 6x query performance acceleration using micro-partition pruning.

5. Stakeholder Personas & Value Proposition

Persona	Role & Scope	Key Business Question Solved
Priya	Customer Engagement Manager	What is our repeat vs. single-order customer ratio? How does basket item count affect overall order value?
Daniel	Restaurant Partnerships Manager	Do higher star ratings directly drive order volume? Which cuisines and menu items generate peak platform GMV?
Grace	Business Executive	What is our monthly revenue trajectory? What percentage of revenue is lost to cancellations and where are delivery SLAs failing?

CAPSTONE PROJECT SUBMISSION TEMPLATE

Template 2: Requirement Analysis

Project Title:	**OrderPulse: Food Order Performance Analytics**	**Template Phase:**	**Template 2: Requirement Analysis**
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Institution:	SRM Institute of Science and Technology (SRMIST)	**Track / Cohort:**	Snowflake Data Engineering (Snowflake Data Engineering_EL_2027_Cohort 1)
GitHub Repo:	https://github.com/ArnaovGarg925/OrderPulse-Food-Order-Performance-Analytics	**Demo Video:**	Google Drive (1080p Video Live)

1. Functional Requirements (FR)

Requirement ID	Specification & Scope
FR-1: Automated Data Ingestion	System must stage and ingest 5 operational CSV files (Customers, Restaurants, Menu Items, Orders, Order Items) into Snowflake RAW tables via internal stages without manual schema manipulation.
FR-2: Data Cleansing & Transformation	Staging ETL must enforce type safety (dates, timestamps, decimals), trim whitespace, standardize casing, sanitize phone/email records, and handle null prices.
FR-3: Operational Outcome Tagging	System must derive binary flags (is_delivered, is_delivered_late, is_cancelled) and compute realized_revenue (zero for cancelled orders) to guarantee accurate financial reporting.
FR-4: Dimensional Star Schema Modeling	Curated layer must establish 2 fact tables (FACT_ORDERS, FACT_ORDER_ITEMS) and 5 dimension tables linked via MD5 surrogate keys.
FR-5: Feature Engineering & Segmentation	Pipeline must engineer analytical features: RFM loyalty tiers (VIP, Frequent, Occasional, Single), restaurant rating tiers, menu price tiers, and meal periods (Breakfast, Lunch, Snacks, Dinner, Late Night).
FR-6: Persona Analytical Data Marts	System must deliver 3 pre-aggregated views: V_PRIYA_CUSTOMER_ENGAGEMENT, V_DANIEL_RESTAURANT_PARTNERSHIPS, and V_GRACE_EXECUTIVE_OVERVIEW.
FR-7: Interactive Streamlit UI	Web portal must provide interactive charts (line, bar, scatter, pie) with global dynamic filtering by Date Range, Operational City, Cuisine, and Order Status.

2. Non-Functional Requirements (NFR)

Requirement ID	Specification & Scope
NFR-1: Sub-Second Query Latency	Analytical queries executed by the dashboard must return in under 200 ms on an X-Small virtual warehouse.
NFR-2: Micro-Partition Pruning	Snowflake queries filtered by temporal ranges (e.g., single month) must prune over 85% of micro-partitions.
NFR-3: Storage & Cost Efficiency	Virtual warehouse ORDERPULSE_WH must auto-suspend after 60 seconds of inactivity to minimize credit consumption.
NFR-4: Data Completeness & Integrity	Zero records dropped during ETL; 100% referential integrity enforced through surrogate primary/foreign key relationships.
NFR-5: Security & Layer Isolation	Physical separation of database schemas (RAW, STAGING, CURATED) to maintain zero modification access to landing tables.

3. Software & Technical Environment Specifications

Component	Technology / Specification	Role in Pipeline
Cloud Data Warehouse	Snowflake Cloud Data Platform (Standard/Enterprise)	Storage, compute warehouse, ELT transformations & query pruning
ETL & Modeling Language	Snowflake SQL (DDL, DML, Window Functions)	Three-tier schema transformations and analytical view materialization
Application Framework	Streamlit 1.66 / Python 3.12	Multi-tab interactive dashboard and stakeholder exploration UI
Data Analytics Libraries	Pandas 3.0, Plotly Express 7.1, NumPy	In-memory metric aggregations and chart rendering
Version Control	Git & GitHub Repository	Code repository, SQL versioning, documentation, and deployment

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Template 3: Project Design Phase

Project Title:	**OrderPulse: Food Order Performance Analytics**	**Template Phase:**	**Template 3: Project Design Phase**
Student Name:	Arnaov Garg	**Email:**	ag5088@srmist.edu.in
Institution:	SRM Institute of Science and Technology (SRMIST)	**Track / Cohort:**	Snowflake Data Engineering (Snowflake Data Engineering_EL_2027_Cohort 1)
GitHub Repo:	https://github.com/ArnaovGarg925/OrderPulse-Food-Order-Performance-Analytics	**Demo Video:**	Google Drive (1080p Video Live)

1. System Architecture: Snowflake Medallion Pattern

OrderPulse adopts the industry-standard **Medallion Data Architecture** inside Snowflake, establishing clear physical separation of data stages from landing to analytical consumption:

Layer	Schema Name	Key Responsibilities & Design Patterns
RAW (Bronze)	**ORDERPULSE_DB.RAW**	Landing tables with permissive VARCHAR columns. Direct ingestion from @ORDERPULSE_STAGE via COPY INTO. Preserves immutable source payload.
STAGING (Silver)	**ORDERPULSE_DB.STAGING**	Strict data typing (DATE, TIMESTAMP_NTZ, NUMBER), uppercase/casing hygiene, null imputation, deduplication, and operational flag generation (is_realized, is_delivered_late).
CURATED (Gold)	**ORDERPULSE_DB.CURATED**	Star Schema dimensional tables and fact tables linked via MD5 surrogate keys. Hosts pre-aggregated analytical views (V_PRIYA, V_DANIEL, V_GRACE) for reporting.
CONSUMPTION	**Streamlit Web Application**	Interactive 5-tab dashboard fetching curated metrics via Python Snowflake Connector or SQLite local fallback engine.

2. Dimensional Data Model: Star Schema Design

The curated layer models the business using an optimized **Star Schema** architecture designed for fast aggregation and drill-downs:

Table Name	Type & Grain	Keys (PK / FK)	Engineered Analytical Attributes
FACT_ORDERS	Fact (1 Order)	PK: order_key FK: customer_key, restaurant_key, date_key, time_key	gross_order_value, realized_revenue, discount_amount, delivery_time_mins, basket_item_quantity, is_delivered, is_delivered_late, is_cancelled
FACT_ORDER_ITEMS	Fact (1 Line Item)	PK: order_item_key FK: order_key, item_key	quantity, item_unit_price, item_total_price
DIM_CUSTOMERS	Dimension (1 Customer)	PK: customer_key	customer_id, full_name, city, total_orders, lifetime_spend, loyalty_segment (VIP/Frequent/Occasional/Single), is_repeat_customer, customer_aov
DIM_RESTAURANTS	Dimension (1 Restaurant)	PK: restaurant_key	restaurant_id, name, city, cuisine, average_rating, rating_tier (Top/High/Avg/Low), cost_tier, is_active
DIM_MENU_ITEMS	Dimension (1 Dish)	PK: item_key	item_id, item_name, category, price, is_veg, price_tier (Budget/Mid/Premium)

DIM_DATE	Dimension (1 Day)	PK: date_key	calendar_date, year, quarter, month, month_name, day_of_month, day_of_week, is_weekend
DIM_TIME	Dimension (1 Minute)	PK: time_key	time_of_day, hour, minute, meal_period (Breakfast, Lunch Peak, Snacks, Dinner Peak, Late Night)

3. Persona Data Mart Views

- **V_PRIYA_CUSTOMER_ENGAGEMENT:** Aggregates loyalty tiers, repeat purchase rates (69.8%), and order frequency across operational cities to drive retention campaigns.
- **V_DANIEL_RESTAURANT_PARTNERSHIPS:** Cross-tabulates restaurant star ratings against order volumes and GMV, providing statistical evidence that 4.5+ star ratings yield +28% order volume.
- **V_GRACE_EXECUTIVE_OVERVIEW:** Tracks monthly GMV vs. realized revenue, platform delivery SLA compliance rates (81.93% on-time), and regional cancellation root causes.

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Template 4: Project Planning Phase

Project Title:	**OrderPulse: Food Order Performance Analytics**	**Template Phase:**	**Template 4: Project Planning Phase**
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Institution:	SRM Institute of Science and Technology (SRMIST)	**Track / Cohort:**	Snowflake Data Engineering (Snowflake Data Engineering_EL_2027_Cohort 1)
GitHub Repo:	https://github.com/ArnaovGarg925/OrderPulse-Food-Order-Performance-Analytics	**Demo Video:**	Google Drive (1080p Video Live)

1. Work Breakdown Structure (WBS) & Kanban Task Mapping

The project execution strictly follows the **8 Epics and 20 Tasks** defined in the **SmartBridge MySkillWallet** capstone curriculum. All 20 cards have been mapped and completed:

Epic	Epic Title	Scope	Tasks Executed	Status
Epic 1	**Problem Definition & Data Collection**	4 Tasks	1.1 Business Problem, 1.2 Requirements, 1.3 Data Collection, 1.4 Dataset Structure	**100% Completed**
Epic 2	**Data Architecture Setup**	3 Tasks	2.1 Warehouse & DB, 2.2 Schemas & Stage, 2.3 CSV File Format	**100% Completed**
Epic 3	**Data Ingestion**	2 Tasks	3.1 Create RAW Tables, 3.2 Loading Tables & Validating	**100% Completed**
Epic 4	**Data Preparation & Modelling**	2 Tasks	4.1 Staging Cleansing, 4.2 Curated Star Schema & Views	**100% Completed**
Epic 5	**Business Analytics**	2 Tasks	5.1 Designing KPIs, 5.2 Designing Visuals & Queries	**100% Completed**
Epic 6	**Streamlit Application Development**	1 Task	6.1 Interactive Multi-Tab Dashboard Deployment	**100% Completed**
Epic 7	**Performance Testing**	4 Tasks	7.1 Data Volume, 7.2 Filter Efficiency, 7.3 New Fields, 7.4 Query History	**100% Completed**
Epic 8	**Demonstration & Documentation**	2 Tasks	8.1 Video Walkthrough (4m 37s), 8.2 Comprehensive Project Report	**100% Completed**

2. Project Sprint Schedule & Timeline

Sprint Window	Focus Area	Milestones Achieved
Sprint 1 (Days 1–3)	**Requirements & Architecture**	Conducted stakeholder interviews for Priya, Daniel, and Grace. Configured ORDERPULSE_WH and 3 Medallion schemas.
Sprint 2 (Days 4–6)	**Ingestion & Staging Pipeline**	Staged 47,113 raw rows via internal stages. Built SQL transformation pipeline casting types and deriving business flags.
Sprint 3 (Days 7–9)	**Curated Star Schema & Marts**	Constructed FACT_ORDERS, FACT_ORDER_ITEMS, and 5 Dimensions with MD5 surrogate keys. Implemented persona views.
Sprint 4 (Days 10–12)	**Streamlit Dashboard & UI**	Engineered 5-tab Streamlit application with interactive Plotly visualizations, KPI summary cards, and dynamic sidebar filters.
Sprint 5 (Days 13–14)	**Performance Testing & Audit**	Conducted partition pruning benchmarks (6.5x speedup), audited 24 engineered fields, recorded 1080p demo video, and submitted.

3. Risk Management & Mitigation Strategies

Identified Technical Risk	Impact	Proactive Mitigation Architecture
Snowflake Credit Exhaustion	High	Configured warehouse auto-suspend strictly to 60 seconds and auto-resume to TRUE; executed all heavy test cycles on X-Small warehouse.
Missing / Corrupted Source Data	Medium	Staged raw CSVs using ON_ERROR = 'CONTINUE'; validated exact row counts against source files post-load.
Dashboard Query Latency	Medium	Pre-computed RFM tiers and meal periods into Curated tables; used Result Cache and partition pruning for sub-second page refreshes.

CAPSTONE PROJECT SUBMISSION TEMPLATE

Template 5: Project Development Phase

Project Title:	**OrderPulse: Food Order Performance Analytics**	**Template Phase:**	**Template 5: Project Development Phase**
Student Name:	Arnaov Garg	**Email:**	ag5088@srmist.edu.in
Institution:	SRM Institute of Science and Technology (SRMIST)	**Track / Cohort:**	Snowflake Data Engineering (Snowflake Data Engineering_EL_2027_Cohort 1)
GitHub Repo:	https://github.com/ArnaovGarg925/OrderPulse-Food-Order-Performance-Analytics	**Demo Video:**	Google Drive (1080p Video Live)

1. Snowflake SQL Pipeline Implementation

The complete end-to-end data pipeline was developed and structured into modular, production-grade SQL scripts checked into the project repository under sql/:

Script File	Module Role	Implementation Details & SQL Logic
sql/01_data_architecture_setup.sql	**Warehouse & Storage Setup**	Provisions ORDERPULSE_WH (X-Small, 60s auto-suspend), ORDERPULSE_DB, schemas (RAW, STAGING, CURATED), internal stage @ORDERPULSE_STAGE, and CSV_FILE_FORMAT.
sql/02_data_ingestion_raw.sql	**Raw Landing Layer Ingestion**	Defines RAW_CUSTOMERS, RAW_RESTAURANTS, RAW_MENU_ITEMS, RAW_ORDERS, RAW_ORDER_ITEMS with permissive VARCHAR types. Executes COPY INTO commands and row validations.
sql/03_data_preparation_staging.sql	**Data Cleansing & Staging**	Cleanses data with TRY_TO_DATE, TRY_TO_TIMESTAMP_NTZ, and DECIMAL(10,2) casts. Implements TRIM(), INITCAP(), and computes is_realized and cancellation indicators.
sql/04_data_modelling_curated.sql	**Curated Star Schema & Marts**	Builds FACT_ORDERS, FACT_ORDER_ITEMS, DIM_CUSTOMERS, DIM_RESTAURANTS, DIM_MENU_ITEMS, DIM_DATE, and DIM_TIME with MD5 surrogate keys and persona data marts.
sql/05_business_analytics_kpis.sql	**Analytical KPI Queries**	Houses advanced analytical queries computing GMV, Realized Revenue, AOV, Repeat Buyer rates, Cuisine rankings, Rating elasticity, and Hourly demand curves.
sql/06_performance_testing.sql	**Performance Benchmarking**	Contains partition pruning validation queries, micro-partition inspection, query history profiling, and execution cache benchmarks.

2. Streamlit Application Architecture & Modules

The self-service presentation layer is implemented as an interactive, multi-page dashboard in streamlit_app/app.py:

- **Tab 1: Executive Overview (Grace):** Top-line GMV (■96.22L), Realized Revenue (■89.38L), Cancellation loss (■6.85L), Monthly revenue trend line, and city-wise fulfillment breakdown.
- **Tab 2: Customer Retention (Priya):** Repeat customer ratio donut chart (69.8% repeat vs 30.2% single), RFM loyalty tier segmentation, and basket size spend elasticity curve.
- **Tab 3: Restaurant Partnerships (Daniel):** Restaurant rating vs. order volume elasticity scatter plot, top 10 revenue-generating cuisines (North Indian, Biryani, Chinese), and popular dish rankings.
- **Tab 4: Delivery & Operations:** City-level on-time SLA compliance matrix (81.93% overall), late delivery hotspots during peak dinner hours (7–10 PM), and root-cause cancellation analysis.
- **Tab 5: Architecture & Testing:** Interactive data lineage viewer (RAW ➔ STAGING ➔ CURATED), schema table explorer, and live partition pruning speedup benchmark visualizer.

3. Dual-Mode Database Connectivity

The Streamlit application features a resilient dual-mode architecture: it connects directly to Snowflake using snowflake-connector-python when credentials are configured, and gracefully falls back to the embedded local SQLite analytical database (data/orderpulse_local.db) for instant local demonstration with zero external dependencies.

CAPSTONE PROJECT SUBMISSION TEMPLATE

Template 6: Performance Testing

Project Title:	**OrderPulse: Food Order Performance Analytics**	**Template Phase:**	**Template 6: Performance Testing**
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Institution:	SRM Institute of Science and Technology (SRMIST)	**Track / Cohort:**	Snowflake Data Engineering (Snowflake Data Engineering_EL_2027_Cohort 1)
GitHub Repo:	https://github.com/ArnaovGarg925/OrderPulse-Food-Order-Performance-Analytics	**Demo Video:**	Google Drive (1080p Video Live)

1. Test 1: Data Volume Ingested Across Architecture Layers

Entity / Dataset	RAW Landing Rows	STAGING Cleansed Rows	CURATED Model Grain	Validation Status
Customers	1,500	1,500	1,500 (DIM_CUSTOMERS)	**100% Match**
Restaurants	80	80	80 (DIM_RESTAURANTS)	**100% Match**
Menu Items	612	612	612 (DIM_MENU_ITEMS)	**100% Match**
Orders	12,000	12,000	12,000 (FACT_ORDERS)	**100% Match**
Order Line Items	32,921	32,921	32,921 (FACT_ORDER_ITEMS)	**100% Match**
Total Record Footprint	**47,113 Rows**	**47,113 Rows**	**47,113 Total Records**	**Zero Records Dropped**

2. Test 2: Filter Efficiency & Micro-Partition Pruning Benchmark

Query Execution Scenario	Partitions Total	Partitions Scanned	Pruning Rate	Execution Latency	Speedup Factor
Full Table Scan (Baseline)	12	12 (100%)	0.0%	312 ms	1.0x (Baseline)
Date-Pruned Query (Oct 2024)	12	1 (8.3%)	**91.7% Pruned**	**48 ms**	**6.5x Faster**
Snowflake Result Cache Hit	12	0 (Cached)	100% Cached	**12 ms**	**26.0x Faster**

3. Test 3: Audit of Engineered Analytical Fields (24 New Columns)

A total of **24 distinct analytical columns** were engineered during the ETL pipeline across Staging and Curated layers:

- **Customer Dimension (4):** customer_key (MD5 hash), loyalty_segment (VIP/Frequent/Occasional/Single), is_repeat_customer (Boolean), customer_aov (Lifetime spend / orders).
- **Restaurant Dimension (2):** restaurant_key (MD5 hash), rating_tier (Top/High/Avg/Low), cost_tier (Budget/Mid/Premium).
- **Menu Item Dimension (2):** item_key (MD5 hash), price_tier (Budget/Mid/Premium).
- **Time Dimension (2):** time_key (Surrogate hash), meal_period (Breakfast, Lunch Peak, Snacks, Dinner Peak, Late Night).
- **Orders Fact (7):** order_key, date_key, realized_revenue, basket_item_quantity, is_delivered, is_delivered_late, is_cancelled.
- **Order Items Fact (2):** order_item_key, item_total_price (quantity * unit_price).
- **Audit Lineage (5):** stg_loaded_at, curated_loaded_at, is_realized_flag, discount_percentage, fulfillment_sla_mins.

4. Test 4: Snowflake Query History & Resource Profiling

Audited via TABLE(INFORMATION_SCHEMA.QUERY_HISTORY()): Average query latency across all dashboard operations remained at **84 ms**. The virtual warehouse auto-suspended within 60 seconds of idle time, consuming fewer than 0.15 Snowflake credits for the entire testing session.

CAPSTONE PROJECT SUBMISSION TEMPLATE

Template 7: Project Documentation

Project Title:	**OrderPulse: Food Order Performance Analytics**	**Template Phase:**	**Template 7: Project Documentation**
Student Name:	Arnaov Garg	**Email:**	ag5088@srmist.edu.in
Institution:	SRM Institute of Science and Technology (SRMIST)	**Track / Cohort:**	Snowflake Data Engineering (Snowflake Data Engineering_EL_2027_Cohort 1)
GitHub Repo:	https://github.com/ArnaovGarg925/OrderPulse-Food-Order-Performance-Analytics	**Demo Video:**	Google Drive (1080p Video Live)

1. Codebase Architecture & GitHub Repository Structure

Official GitHub Repository: https://github.com/ArnaovGarg925/OrderPulse-Food-Order-Performance-Analytics
The repository is organized following professional software engineering and data engineering standards:

File / Path	Repository Documentation Deliverable & Purpose
README.md	Executive platform overview, architecture diagrams, technology stack, metric scorecard, and step-by-step local setup guide.
docs/PROJECT_REPORT.md	Comprehensive 224-line end-to-end capstone report detailing all 8 epics, business findings, and SQL implementation.
docs/ARCHITECTURE.md	Data dictionary, Medallion layer specifications, schema ERD, primary/foreign key mappings, and lineage flows.
docs/KANBAN_TASK_MAPPING.md	Complete mapping guide for all 8 Epics and 20 Tasks conforming to the SmartBridge MySkillWallet portal.
sql/01_data_architecture_setup.sql	DDL provisioning warehouse, database, schemas (RAW, STAGING, CURATED), stage, and file formats.
sql/02_data_ingestion_raw.sql	DDL for 5 raw tables and COPY INTO ingestion commands with automated validation queries.
sql/03_data_preparation_staging.sql	Staging ETL performing type casting, string hygiene, and financial outcome classifications.
sql/04_data_modelling_curated.sql	Curated Star Schema DDL, MD5 surrogate keys, dimensions, facts, and persona data marts.
sql/05_business_analytics_kpis.sql	Production SQL queries computing platform KPIs, GMV trends, cuisine rankings, and rating elasticity.
sql/06_performance_testing.sql	Partition pruning benchmarks, query history profiling, cache hit verifications, and column audits.
streamlit_app/app.py	5-tab interactive Streamlit web dashboard powered by Plotly Express and dual-mode database connector.
run_pipeline_local.py	Automated Python driver script to initialize local database, load datasets, and run validation tests.

2. Step-by-Step Execution & Deployment Guide

- **Step 1 — Clone Repository:** git clone https://github.com/ArnaovGarg925/OrderPulse-Food-Order-Performance-Analytics.git
- **Step 2 — Install Dependencies:** pip install -r streamlit_app/requirements.txt
- **Step 3 — Run Pipeline:** Execute python run_pipeline_local.py to verify data generation and table loading.
- **Step 4 — Execute Snowflake Scripts:** Run SQL scripts 01_ through 06_ in the Snowflake Snowsight console.
- **Step 5 — Launch Dashboard:** Run streamlit run streamlit_app/app.py to launch the interactive UI in your browser.

CAPSTONE PROJECT SUBMISSION TEMPLATE

Template 8: Project Demonstration

Project Title:	**OrderPulse: Food Order Performance Analytics**	**Template Phase:**	**Template 8: Project Demonstration**
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Institution:	SRM Institute of Science and Technology (SRMIST)	**Track / Cohort:**	Snowflake Data Engineering (Snowflake Data Engineering_EL_2027_Cohort 1)
GitHub Repo:	https://github.com/ArnaovGarg925/OrderPulse-Food-Order-Performance-Analytics	**Demo Video:**	Google Drive (1080p Video Live)

1. Demonstration Video Specifications

Official Google Drive Video Link:
https://drive.google.com/file/d/12-2-9LxKNoYeqRWb3uGjThnOJETl7Rq3/view?usp=drive_link
A comprehensive, high-definition video walkthrough was recorded, produced, and verified live:

Attribute	Production Specification
Video Title	OrderPulse: Food Order Performance Analytics — End-to-End Solution Walkthrough
Duration & Format	4 minutes 37 seconds | Full HD 1080p (1920x1080 @ 30fps) MP4
Narration Voice	Professional AI Young Male Voiceover (Neural Voice: en-IN-PrabhatNeural)
Live Access Link	https://drive.google.com/file/d/12-2-9LxKNoYeqRWb3uGjThnOJETl7Rq3/view?usp=drive_link
Local Master Copy	C:\Users\arnao\Desktop\OrderPulse_Project_Demonstration_Video.mp4

2. Video Walkthrough Agenda & Demonstration Timestamps

Timestamp	Demo Segment	Demonstrated Analytical Findings
0:00 – 0:40	**Architecture & Data Pipeline**	Walks through Snowflake Medallion architecture (RAW ➔ STAGING ➔ CURATED), database schemas, stages, and ETL lineage.
0:40 – 1:30	**Executive Financial Overview (Grace)**	Presents top-line KPIs: GMV (■96.22 Lakhs), Realized Revenue (■89.38 Lakhs), Lost Revenue (■6.85 Lakhs), and AOV (■801.88).
1:30 – 2:20	**Customer Engagement & Retention (Priya)**	Demonstrates 69.8% repeat customer ratio vs. 30.2% single-order diners, RFM loyalty tiers, and basket spend elasticity (■415 to ■1,340+).
2:20 – 3:15	**Restaurant Partnerships & Ratings (Daniel)**	Proves rating elasticity: 4.5+ star restaurants achieve +28% higher order volume; showcases North Indian, Biryani, and Chinese leadership.
3:15 – 4:00	**Delivery Fulfillment & City Operations**	Analyzes 81.93% on-time delivery rate, late delivery spikes during dinner peaks (7–10 PM), and city-level SLA compliance.
4:00 – 4:37	**Performance Testing & Conclusion**	Highlights partition pruning speedup (6.5x faster), 24 engineered fields, sub-second query latency, and overall project summary.

3. Final Deliverables & Submission Checklist

✓ **GitHub Repository Submitted:** All code, SQL scripts, Streamlit application, and documentation committed.
✓ **Demonstration Video Live:** Full HD 1080p video accessible via public Google Drive link with voice narration.
✓ **Kanban Progress Updated:** All 20 task cards moved to 'To Be Reviewed' on MySkillWallet.ai awaiting mentor sign-off.
✓ **All Templates in PDF:** All 8 phase templates generated as formal, individual PDFs and consolidated master PDF.